GENESIEVE: A KNOWLEDGE NETWORK INTERFACE FOR GENE DISCOVERY ACROSS DIVERSE CROP SPECIES

by

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Presented to the Faculty of the Graduate School of

The University of Texas at Arlington in Partial Fulfillment

of the Requirements

for the Degree of

MASTERS OF SCIENCE

August 2026

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ACKNOWLEDGEMENTS

I would like to express my heartfelt gratitude to my supervising professor Dr. Chengkai Li, for his constant guidance, support, and patience throughout my master's studies. Working with him has been an inspiring and rewarding experience. His support and advice enabled me to achieve things in life I had only ever dreamt of. I really appreciate his kindness and encouragement. I will always remember the chess games we played together.

I would also like to thank my committee members, Dr. Justin Vaughn, Dr. Gautam Das and Dr. Negin Fraidouni, for their time and thoughtful feedback. Their suggestions were crucial for me to successfully complete my research. They helped me improve the quality of my work and guided me to look at my research from new points of view. I am grateful for the effort they put into my thesis and sharing their knowledge with me.

I am very thankful to all my labmates. They made every day in the lab enjoyable and meaningful. I am happy for all the discussions we had, both about research and about everyday life. I also enjoyed our chess games, mahjong sessions, and fun conversations. Being part of such a friendly group made my time at the university a great experience.

I also want to thank the faculty and staff in the Computer Science and Engineering Department for their help and support. Their guidance and organization made it possible for me to focus on my studies and complete my research successfully.

Lastly, I would like to thank my family and friends for their love, care, and constant encouragement. They always believed in me and motivated me to keep going even when things were hard. I dedicate this work to them with deep appreciation.

July 28, 2026

ABSTRACT

GENESIEVE: A KNOWLEDGE NETWORK INTERFACE FOR GENE DISCOVERY ACROSS DIVERSE CROP SPECIES

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Identifying genes responsible for plant traits remains a fundamental challenge in plant biology. Although advances in genomics have enabled researchers to identify large sets of candidate genes associated with a trait, experimentally validating each candidate remains costly and time-consuming, creating a need for computational methods that can prioritize likely causal genes while providing interpretable biological evidence. We present GeneSieve, a system for candidate gene prioritization and gene discovery across diverse plant species. Given a natural-language trait description and a set of candidate genes from any angiosperm species, GeneSieve constructs a Trait–Gene Association Graph (TGAG) for each candidate by integrating four complementary evidences: trait similarity, trait–gene associations, gene co-expression, and sequence homology. The underlying biological data comes from four keystone plant species—rice, maize, soybean, and Arabidopsis—linking 1,055 traits to 127,611 genes through 952,630 curated gene–trait associations and more than 2.4 billion gene–gene co-expression relationships. Candidate genes are prioritized using a relational graph convolutional network (R-GCN) that learns from TGAGs' heterogeneous topol-

ogy and relationships. The model was trained and evaluated using cross-validation on more than 3,000 TGAGs generated from a curated benchmark of experimentally validated causal gene–trait associations. GeneSieve consistently outperformed baseline methods, placing the validated causal gene among the top five candidates in approximately 70% of queries. GeneSieve also provides an interactive web platform for candidate gene prioritization and evidence visualization. The novelty of GeneSieve lies in addressing key limitations of existing candidate gene prioritization methods by constructing query-specific heterogeneous graphs that integrate LLM-derived trait similarity, curated trait–gene associations, RNA-seq-derived gene co-expression, and sequence homology into a unified graph framework. This enables evidence-driven R-GCN-based prioritization across diverse monocot and dicot plant species, rather than relying on static knowledge graphs, isolated evidence sources, or direct LLM prediction.

TABLE OF CONTENTS

ACKNOWLEDGEMENTS iii

ABSTRACT v

LIST OF ILLUSTRATIONS ix

LIST OF TABLES xi

Chapter Page

1. Introduction 1

2. LITERATURE SURVEY 5

2.1 Overview 5

2.2 Candidate Gene Prioritization 6

2.3 Machine Learning for Candidate Gene Prioritization 7

2.4 Graph Learning and Knowledge Graphs for Biological Discovery . . . 8

2.5 Large Language Models in Computational Biology 10

2.6 Research Gaps and Contributions 12

3. Materials and Methodology 13

3.1 Database and Terminology 13

3.2 System Backend and TGAG Construction 14

3.3 Gene ranking models 19

3.3.1 Relational-graph convolutional network 20

3.3.2 SVD 25

3.4 Trait Similarity Models 26

4. Results and Discussion 29

4.1 Relational-Graph Convolutional Network (R-GCN) 31

4.2 SVD 32
4.3 Baseline Models 33
4.4 Trait Similarity Models 35
5. CONCLUSION AND FUTURE WORK 39
5.1 Summary of Contributions 39
5.2 Future Work 40
Appendix
REFERENCES 41
BIOGRAPHICAL STATEMENT 44

LIST OF ILLUSTRATIONS

Figure Page

3.1 GeneSieve **System Backend Workflow.** Generating TGAGs using user-provided inputs and known tables 15

3.2 Trait–Gene Association Graph Construction 16

3.3 Example TGAG 19

3.4 R-GCN V1 Architecture 22

3.5 R-GCN V2 Architecture 23

4.1 **Cumulative Comparison Plot:** Rank is calculated considering folder length = 20 30

4.2 **Cumulative Comparison Plot:** Rank is calculated considering folder length = # of Candidate Genes 30

4.3 **Non Cumulative Comparison Plot:** Rank is calculated considering folder length = 20 31

4.4 **Non Cumulative Comparison Plot:** Rank is calculated considering folder length = # of Candidate Genes 31

4.5 **Calculation of the score for a TGAG.** Using summation of products along unique paths connecting the Input Trait and Candidate Gene . . 34

4.6 **Doc2Vec score normalization.** Original Doc2Vec scores plotted against their normalized scores. 35

4.7 **Human expert score normalization.** Original human expert annotation scores plotted against their normalized scores. 36

4.8 **Fine-tuned GPT-4o score normalization.** Original fine-tuned GPT-4o scores plotted against their normalized scores. 36
4.9 **Score Comparison** Doc2Vec (stretched) vs Human Expert Score . . 37
4.10 **Score Comparison** Doc2Vec (Normalized) vs Human Expert Score (Normalized) 37
4.11 **Score Comparison** FT-GPT Score vs Human Expert Score 38

LIST OF TABLES

Table Page

3.1 Model hyperparameters and evaluated search space. 21

3.2 Comparison of the two R-GCN architectures. 24

3.3 Cumulative error across 5 folds using pct and norm scores. 28

CHAPTER 1

Introduction

Identifying genes responsible for plant traits remains a fundamental challenge in plant biology. Although advances in genomics have enabled researchers to generate large sets of candidate genes, experimentally validating each candidate is costly and time-consuming, creating a need for computational approaches that can prioritize promising genes while providing interpretable biological evidence.

To address this challenge, we present GeneSieve, a system for candidate gene prioritization and gene discovery across diverse dicot and monocot plant species. The system derives evidence-supported relationships from biological data collected across four "keystone" plant species—rice, maize, soybean, and Arabidopsis. Its central data model is the Trait–Gene Association Graph (TGAG), which integrates four types of relationships: trait–trait similarity, trait–gene associations, gene co-expression, and gene homology. Of these, trait–gene associations and gene co-expression constitute TGAG's foundational biological evidence, represented respectively by quantitative trait locus (QTL) associations and RNA-seq-derived co-expression data. Because most genetic studies associate a genomic region with a trait without identifying the specific causal genes within that region, we assemble a large collection of QTL associations from the four keystone species and link each trait to the genes located in its associated genomic regions. The resulting gene–trait (QTL) table links 1,055 traits to 127,611 genes, producing 952,630 distinct gene–trait associations. Building on this database, we assess each represented gene for co-expression with all other genes in the corresponding genome using more than 750 RNA-seq experiments available

for each keystone species, producing over 2.4 billion distinct gene–gene co-expression relationships.

Given a query, consisting of a natural-language trait description and candidate genes from any angiosperm species, GeneSieve constructs a TGAG for each candidate. A fine-tuned large language model (LLM) matches the queried trait to semantically similar traits in the QTL table. Through the QTL associations of these matched traits, the system identifies genes located in the corresponding trait-associated genomic regions. In parallel, GeneSieve uses the Basic Local Alignment Search Tool (BLAST) [1] to compare the protein sequence of the candidate gene against the protein sequences of genes from the four keystone species and thereby identify homologous genes. The system then retrieves the genes co-expressed with these homologs and determines where this co-expression-derived gene set intersects with the genes associated with the matched traits through QTL relationships. In other words, an evidence path is formed when a QTL-associated gene for a matched trait is also co-expressed with a homolog of the candidate gene.

To prioritize candidate genes, GeneSieve employs a relational graph convolutional network (R-GCN) that learns representations from the topology and relationship features of each TGAG and assigns each candidate a relevance score, which is then used to rank the candidates. The model is trained and evaluated using cross-validation on more than 3,000 TGAGs generated from a curated benchmark of experimentally validated cases in which a single causal gene—rather than only a broad genomic region—has been conclusively shown to influence a plant trait. The proposed custom R-GCN consistently outperforms three baseline ranking methods: singular value decomposition (SVD), random ranking, and a sum-of-products algorithm. SVD is a matrix factorization technique that decomposes the gene–trait association structure of each TGAG into lower-dimensional latent factors, which are then used to

estimate the relevance of candidate genes. Random ranking assigns candidates random rankings, while the sum-of-products method estimates candidate relevance by multiplying the relationship weights along each unique path connecting the queried trait to a candidate gene and summing these products across all such paths. The R-GCN successfully ranks the experimentally validated causal gene among the top three candidates in over 50% of queries and among the top five candidates in approximately 75% of queries.

GeneSieve provides an interactive D3.js web interface at https://idir.uta.edu/genesieve/, where plant scientists can submit a natural-language trait description and a FASTA file containing candidate gene sequences, obtain a ranked list of candidates, and explore interactive TGAG visualizations that connect the queried trait to each candidate gene. The interface also provides access to TGAGs generated for the curated benchmark. A demonstration video is available at https://vimeo.com/1070857678. The GeneSieve codebase, including the experimental scripts and results, is publicly available at https://github.com/idirlab/genesieve.

Recent studies have demonstrated the effectiveness of heterogeneous graph learning for candidate gene prioritization, but they generally leverage static phenotype-gene knowledge graphs in the human disease domain [2] and do not construct query-specific biological graphs from heterogeneous evidence sources for plants. Likewise, RNA-seq-derived co-expression networks have been widely used for gene function prediction [3], yet existing approaches typically analyze co-expression as a standalone network rather than dynamically integrating query-specific co-expression relationships with multiple other evidence sources into a unified graph representation. Existing applications of large language models in gene prioritization primarily treat the LLM as a direct prediction or ranking engine [4], and do not leverage LLMs to infer semantic similarity between natural language plant traits and curated trait annotations for

graph construction. Furthermore, most graph-based gene prioritization methods are designed for human genetics or individual plant species, limiting their applicability across diverse crop genomes.

GeneSieve's novelty and contributions lie in addressing these gaps by constructing query-specific TGAGs that integrate LLM-derived trait similarity, curated trait–gene associations, RNA-seq-derived co-expression, and sequence homology into a unified graph representation, which is then ranked using an R-GCN model to prioritize candidate genes across both dicot and monocot plant species. This aligns with one of the overarching visions of the United States Department of Agriculture (USDA)—to facilitate agricultural research initiatives and to address agricultural challenges. The USDA intends to achieve this through comprehensive data generation and computational analysis of genomic data—principles that are central to GeneSieve.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview

The identification of genes responsible for complex plant traits has long been a central challenge in plant genetics. Advances in genomics and high-throughput sequencing have generated vast amounts of biological data, including quantitative trait loci (QTLs), transcriptomic datasets, protein sequences, and comparative genomics resources such as Gramene [5]. These datasets have enabled the development of computational approaches that prioritize candidate genes for experimental validation, reducing the cost and effort associated with traditional gene discovery. Over time, candidate gene prioritization methods have evolved from approaches based on individual evidence sources, such as sequence homology and gene co-expression, to machine learning and graph-based techniques capable of integrating increasingly complex biological relationships.

More recently, graph neural networks and large language models (LLMs) have introduced new opportunities for integrating heterogeneous biological data and interpreting natural-language trait descriptions. However, existing approaches remain limited by their reliance on static biological networks, single evidence sources, or species-specific datasets, and often provide limited biological interpretability. This chapter reviews the major computational approaches to candidate gene prioritization, including traditional bioinformatics methods, machine learning and graph-based models, and emerging applications of LLMs in computational biology. It concludes by identifying the remaining research gaps that motivate the development of GeneSieve.

2.2 Candidate Gene Prioritization

Candidate gene prioritization aims to identify the genes most likely to influence a phenotypic trait from a larger set of candidate genes. Traditional genetic studies typically employ quantitative trait locus (QTL) mapping or genome-wide association studies (GWAS) to associate genomic regions with traits. However, because these regions often encompass dozens or hundreds of genes, additional computational methods are needed to prioritize candidates for experimental validation.

Early computational approaches relied on individual sources of biological evidence. Sequence homology has long been used to infer gene function based on evolutionary conservation, with tools such as the Basic Local Alignment Search Tool (BLAST) enabling the identification of homologous genes across species [6, 7]. Comparative genomics resources such as Gramene further support cross-species analyses by integrating genomic and functional information from multiple plant species [5]. Similarly, curated QTL databases, including Q-TARO [8], provide experimentally validated associations between genomic regions and phenotypic traits, allowing researchers to identify genes located within trait-associated intervals. Gene co-expression networks derived from RNA-seq experiments provide an additional source of functional evidence by identifying genes with coordinated expression patterns that are likely to participate in related biological processes.

Although these approaches have significantly advanced candidate gene discovery, each relies on a single source of biological evidence and therefore captures only part of the complex mechanisms underlying plant traits. Sequence homology cannot account for species-specific functional divergence, QTL-based approaches identify genomic regions rather than causal genes, and co-expression networks often contain indirect or context-dependent relationships. These limitations have motivated the

development of computational methods capable of integrating multiple complementary evidence sources to improve candidate gene prioritization.

2.3 Machine Learning for Candidate Gene Prioritization

- **Machine learning approaches for integrating biological evidence.** The increasing availability of large-scale genomic and transcriptomic datasets has driven the adoption of machine learning techniques for candidate gene prioritization. Unlike traditional approaches that rely on a single source of biological evidence, machine learning models can learn complex relationships from heterogeneous datasets and identify patterns that are difficult to capture using rule-based or statistical methods alone. Early applications employed supervised learning algorithms such as support vector machines, random forests, and Bayesian models, using handcrafted biological features derived from gene expression, sequence similarity, functional annotations, and protein interactions to distinguish causal genes from non-causal candidates.
- **Deep learning and graph-based representation of biological networks.** Recent advances in deep learning have enabled models to learn informative representations directly from biological data without extensive feature engineering. In particular, graph neural networks (GNNs) have emerged as a powerful framework for modeling biological systems, where genes, traits, proteins, and other biological entities can naturally be represented as nodes connected through diverse biological relationships. By propagating information across graph neighborhoods, GNNs capture both local and global network structure, making them well suited for candidate gene prioritization.
- **Heterogeneous graph learning for gene prioritization.** Several recent studies have demonstrated the effectiveness of heterogeneous graph learning for

gene prioritization. For example, Het2Gene constructs heterogeneous phenotype–gene graphs and learns node embeddings to prioritize disease-associated genes, outperforming several existing graph-based methods in the human genetics domain [2]. Similar graph-based approaches have shown that integrating multiple biological relationship types generally improves predictive performance compared with methods relying on a single evidence source.

- **Limitations of existing machine learning approaches.** Despite these advances, several limitations remain. Most graph-based methods have been developed for human disease gene prioritization rather than plant genomics, limiting their applicability to diverse crop species. Furthermore, existing approaches typically construct static biological graphs that are shared across all prediction tasks, rather than dynamically assembling query-specific graphs tailored to an individual trait and candidate gene set. While graph neural networks produce highly accurate rankings, they often provide limited biological interpretability, as predictions are derived from learned node embeddings rather than explicit biological evidence. These limitations motivate the development of frameworks that combine heterogeneous biological evidence within interpretable, query-specific graph representations for plant candidate gene prioritization.

2.4 Graph Learning and Knowledge Graphs for Biological Discovery

Biological systems involve complex interactions among genes, traits, proteins, and molecular processes. As large-scale biological datasets continue to expand, graph-based approaches have emerged as effective frameworks for integrating heterogeneous biological evidence. Knowledge graphs represent biological entities as nodes and their relationships as edges, enabling computational models to capture complex relationships that are difficult to identify using traditional approaches. Such representations

are particularly valuable for candidate gene prioritization because they allow multiple sources of biological information to be integrated within a unified framework.

- **Heterogeneous biological knowledge graphs.** Biological knowledge graphs can integrate diverse evidence sources, including gene–trait associations, gene interactions, functional annotations, and expression-based relationships. By representing multiple entity and relationship types simultaneously, heterogeneous graphs allow models to leverage complementary biological evidence rather than analyzing individual data sources independently. This enables the identification of meaningful associations between genes and phenotypic traits.
- **Graph neural networks for biological graphs.** Graph neural networks (GNNs) have emerged as effective methods for learning representations from biological networks. By aggregating information from neighboring nodes, GNNs capture both network structure and biological context, supporting prediction and ranking of candidate genes. Recent studies have demonstrated the effectiveness of heterogeneous graph learning for gene prioritization by integrating diverse biological relationships into a unified graph representation.
- **Limitations of existing graph-based gene prioritization.** Despite these advances, existing graph-based approaches have primarily focused on human disease gene prioritization and commonly rely on large, static knowledge graphs shared across prediction tasks. Such frameworks are not generally designed for plant genomics, where candidate gene discovery requires integration across diverse crop species and trait-specific biological evidence. Furthermore, GNN predictions can be difficult to interpret because they rely on learned node representations rather than explicit biological reasoning. These limitations motivate query-specific and interpretable graph frameworks for plant candidate gene prioritization.

2.5 Large Language Models in Computational Biology

The rapid advancement of large language models (LLMs) has introduced new opportunities for processing and interpreting complex biological information. Unlike traditional computational approaches that rely on structured databases or manually defined features, LLMs can extract semantic relationships from large-scale textual resources, including scientific literature, biological databases, and functional descriptions. These capabilities are particularly valuable for biological discovery tasks where important relationships are often described using diverse terminology and are difficult to capture through exact matching or conventional computational methods [9, 10].

- **LLMs for biological knowledge extraction and representation.** LLMs have demonstrated the ability to capture biological concepts and relationships from unstructured scientific text, enabling applications such as biomedical knowledge extraction, literature-based discovery, and biological entity representation. Biomedical language models such as BioBERT and PubMedBERT have shown that domain-specific pretraining improves the extraction of biological entities and relationships from scientific literature [11, 12]. More generally, transformer-based architectures learn contextual representations that allow models to capture relationships between terms beyond exact lexical similarity [13].
- **LLMs for phenotype and trait interpretation.** A major challenge in candidate gene prioritization is translating complex phenotype descriptions into meaningful biological representations. Traditional approaches often depend on manually curated ontology terms or keyword-based matching, which can fail when traits are described using different terminology across studies or species. Recent studies have demonstrated that LLMs can improve phenotype interpretation by capturing semantic relationships between phenotype descriptions and biological concepts. For example, LLM-based approaches have been eval-

uated for phenotype-driven gene prioritization, demonstrating their ability to associate natural-language phenotype descriptions with candidate genes [14].

- **LLMs for gene prioritization and biological discovery.** Recent studies have explored the use of LLMs for biological discovery by combining language-based representations with structured biological resources. GeneGPT demonstrated that augmenting LLMs with access to biomedical databases can improve their ability to retrieve and reason over biological information [10]. Additionally, biological foundation models have shown that transformer architectures can learn meaningful representations from large-scale biological sequences and datasets, enabling downstream biological prediction tasks [15, 16]. These studies suggest that LLMs can serve as effective tools for bridging natural language descriptions of biological traits with structured genomic information.
- **Limitations of current LLM-based biological approaches.** Despite their potential, LLM-based methods face several limitations for biological discovery. LLMs primarily learn statistical relationships from training data and may not directly incorporate explicit biological evidence such as gene expression patterns, evolutionary relationships, or functional interactions. Furthermore, most existing LLM-based biological applications have focused on human biomedical datasets, while applications involving plant traits and cross-species candidate gene prioritization remain comparatively limited [9, 10]. Therefore, integrating LLM-derived semantic information with structured biological networks represents an important direction for developing interpretable and evidence-driven gene prioritization frameworks.

2.6 Research Gaps and Contributions

Despite significant advances in candidate gene prioritization through bioinformatics, machine learning, graph learning, and large language models, several limitations remain unresolved. Existing approaches often rely on individual evidence sources or static biological networks that do not adapt to specific trait queries. While graph-based methods can integrate heterogeneous biological relationships, many approaches are primarily designed for human disease applications and provide limited interpretability due to their reliance on learned representations. Similarly, although large language models have demonstrated strong capabilities in understanding biological text and phenotype descriptions, their use in plant genomics and cross-species candidate gene prioritization remains limited. These gaps highlight the need for a unified framework capable of integrating diverse biological evidence, interpreting natural-language trait descriptions, and providing biologically meaningful candidate gene rankings.

This thesis addresses these challenges through the development of GeneSieve, a computational framework that combines large language models, heterogeneous biological evidence integration, and graph neural network-based ranking for cross-species candidate gene prioritization. GeneSieve introduces query-specific Trait–Gene Association Graphs (TGAGs) that integrate LLM-based trait similarity, curated trait–gene associations, gene homology, and RNA-seq-derived co-expression relationships into a unified graph representation. A relational graph convolutional network (R-GCN) is then applied to prioritize candidate genes by learning complex relationships among traits and genes. By combining semantic understanding from LLMs with explicit biological evidence and graph-based learning, GeneSieve provides an interpretable and scalable approach for accelerating gene discovery across diverse plant species.

CHAPTER 3

Materials and Methodology

3.1 Database and Terminology

This section explains various datasets utilized in GeneSieve, as well as key technical terms used throughout this thesis. These datasets contain information pertaining to four keystone plant species—rice, maize, soy, and Arabidopsis. Data for these plants were chosen since they capture broad evolutionary and genomic diversity across flowering plants, providing the ability to generalize GeneSieve to many plant species.

1. QTL (Quantitative Trait Locus) refers to a genomic region that has been identified through genetic analysis as being associated with variation in a measured or observed trait. These traits are alternatively referred to as "Phenotypes". The term originates from QTL mapping, which was developed primarily to study quantitative traits whose variation is influenced by multiple genetic loci and environmental factors. However, the traits represented in the QTL Table are not restricted to strictly quantitative traits, as QTL studies and biological databases may associate genomic regions with a broad range of phenotypic characteristics. Numerous sources of such genomic region–trait associations were accumulated and stored in a dataset that we refer to as the QTL Table. This table contains 952,630 distinct rows of trait-gene pairs, mapping 1,055 traits to 127,611 genes across 4 plant species — rice,[1] maize,[2] soy,[3] and arabidopsis.[4]

[1]Rice Genes: QTARO Database

[2]Maize Genes: Gramene Directory, Gramene ID Mapper Link

[3]Soy Genes: JGI Portal

[4]Arabidopsis Genes: AraGWAS Catalog

Each trait-gene pair, referred to as a *QTL relationship* is associated with a correlation strength. .

2. Co-expression refers to the correlation between the expression levels of two genes. A positive co-expression relationship indicates that the genes tend to be expressed together, while a negative relationship indicates that when one gene is expressed, the other tends not to be. The Co−expression DB, was constructed by comparing gene expression profiles across over 750 RNA-seq experiments available for each of the four keystone plant species. These comparisons identified correlations between pairs of genes, resulting in 2.4 billion distinct gene-gene co-expression relationships. The Co − expression DB is structured as four genome-specific tables. Together, these tables contain 161,721 unique genes in total, comprising the 127,611 genes found in the QTL Table alongside 34,110 additional genes.

Although the QTL and co-expression datasets share approximately 70% of their genes, each contains genes absent from the other. Consequently, the two forms of evidence extend one another reciprocally: QTL associations provide trait-related evidence for genes reached through co-expression, while co-expression relationships connect QTL-associated genes to additional genes not directly represented in the QTL data.

3. GeneSieve also contains a benchmark dataset of 267 folders, each consisting of an input trait description and a list of candidate genes, provided by USDA-ARS. The TGAGs for each candidate gene in these folders are pre-computed. This benchmark dataset is used to train, evaluate, and compare the various gene ranking models.

3.2 System Backend and TGAG Construction

The GeneSieve backend consists of interconnected components, as shown in Fig 3.1. It receives an input trait description and an input FASTA file (containing a

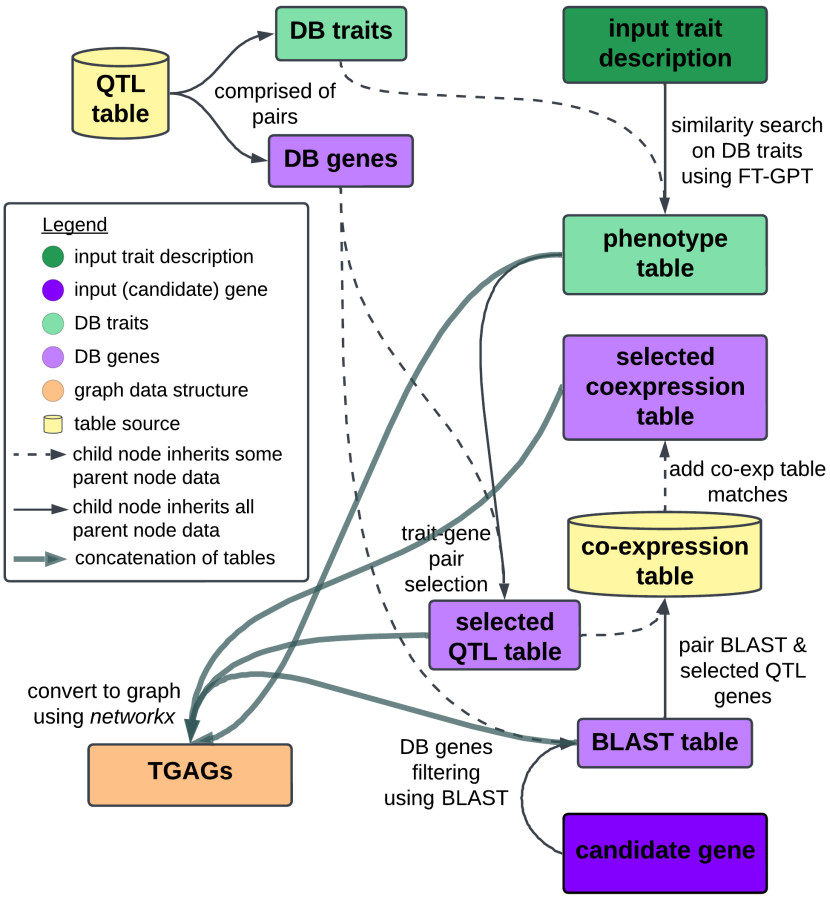


Figure 3.1: GeneSieve **System Backend Workflow.** Generating TGAGs using user-provided inputs and known tables

list of candidate genes) from the user. GeneSieve uses these inputs, and also searches and filters through the known tables, to find relationships between traits and genes. This generates the TGAGs. The following steps detail this process:

A TGAG connects the input trait to the candidate gene with multiple pathways through several layers of traits and genes in between. These pathways consist of various trait-trait (phenotype), trait-gene (QTL) and gene-gene (homology and co-expression) connections, each of which is computed through a dedicated process and integrated sequentially into the graph.

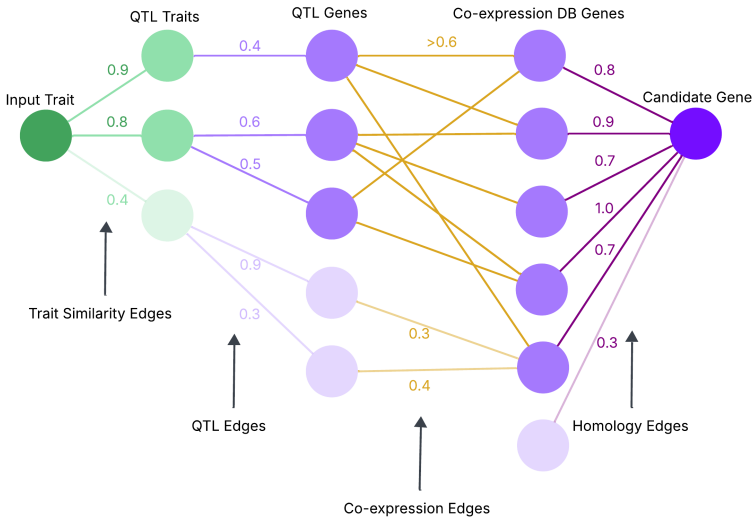


Figure 3.2: Trait–Gene Association Graph Construction

1. Trait similarity: GeneSieve calculates a similarity score between the input trait and each trait from the QTL Table using a fine-tuned GPT-4o model (see Trait Similarity Models). A phenotype list is generated by first identifying all pairs of the input trait and QTL traits (traits from the QTL Table) that have a similarity score exceeding 0.6. From this resulting distribution of valid pairs, the QTL traits containing the top 10% of similarity scores are then selected to form the final list of traits, creating the green connections between the input trait and QTL traits seen in the left-most part of Figure 3.2. The lighter colored connection having weight 0.4 indicates a QTL trait that did not satisfy the trait similarity threshold of 0.6 and hence is not added to the TGAG.

$$Pheno_t = \text{Top}_{10\%} \left\{ t_i \mid t_i \in \text{QTL Table},\ s(t_i) > 0.6 \right\},$$

where $s(t_i)$ is the similarity score between the input trait and the QTL trait t_i, and t is the input trait.

2. *QTL Association*: GeneSieve generates the selected QTL table, which consists of any trait-gene pairs in the QTL table (see Database and Terminology) that include a trait from the phenotype list. This forms the lavender-colored connections to the right side of the trait similarity connections seen in Figure 3.2. The lighter colored connections depict those QTL connections that could not be added into the graph since they are either connected to a trait that failed to satisfy the trait similarity threshold.

$$\text{Selected QTL Table} = \{\, (t, g) \in \text{QTL Table} \mid t \in \text{Pheno Table} \,\}.$$

3. Homology: In GeneSieve, a dataset comprising millions of gene protein sequences is stored in the standard FASTA [17] format. Homology is computed using the Basic Local Alignment Search Tool (BLAST). By calculating sequence similarity scores, this functionality enables researchers to input a protein sequence from any angiosperm species and efficiently identify its homologs within the GeneSieve Co−expression DB. These connections are depicted in the right most side of Figure 3.2, connecting the purple node (candidate gene) to various genes from the Co−expression DB. It can be seen that connections having homology score under the threshold of 0.6 do not make it into the TGAG, depicted by the lighter colored connection with a weight of 0.3.

Pairs with scores above 0.6 are stored in the BLAST hits list.

$$BLAST_c = \{\, (c, g) \in C \times C_{\text{db}} \mid h(c, g) > 0.6 \,\},$$

where c is a candidate gene, C_{db} denotes all genes in Co−expression DB, and $h(c, g)$ is the BLAST homology score.

4. Co-expression: GeneSieve constructs a selected co-expression table by linking genes from the BLAST hits list with genes from the selected QTL table that are positively co-expressed with the DB genes in the BLAST hits list. A gene pair – one from

the BLAST hits list and the other from the selected QTL table — is included in the selected co-expression table if both genes belong to the same species and have a co-expression score greater than 0.6 according to the co-expression table (see Database and Terminology).

$$B \times Q = \{\, (a, b) \mid a \in B, b \in Q \,\}$$

$$\mathrm{S} = \{\, (a, b) \in B \times Q \mid [(a, b) \in C \wedge w(a, b) > 0.6] \vee [(b, a) \in C \wedge w(b, a) > 0.6] \,\},$$

where B is the set of genes in the BLAST hits list, Q is the set of genes in the selected QTL table, and $w(\cdot, \cdot)$ is the co-expression score from CoexpDB.

Finally, to consolidate all evidence relevant to constructing a TGAG (when one exists) for each candidate gene, the phenotype list, selected QTL table, selected co−expression table and the BLAST hits list are concatenated to form the ResultsTable. This table contains the complete set of evidence describing relationships between the input trait description and every candidate gene in the input FASTA file. The ResultsTable is then parsed row by row, where each row represents a single relationship (phenotype similarity, QTL association, coexpression, or homology). Each relationship is added to a global graph containing every connection that forms part of a complete pathway between the input trait and any candidate gene. When a user selects a candidate gene in the interface, the relevant subgraph containing all complete paths from the input trait to the selected candidate gene is extracted from this global graph. This extracted subgraph constitutes the TGAG displayed to the user. The TGAG can subsequently be filtered using the weight threshold sliders shown in Figure 3.3, allowing users to focus on the strongest trait–gene and gene–gene relationships.

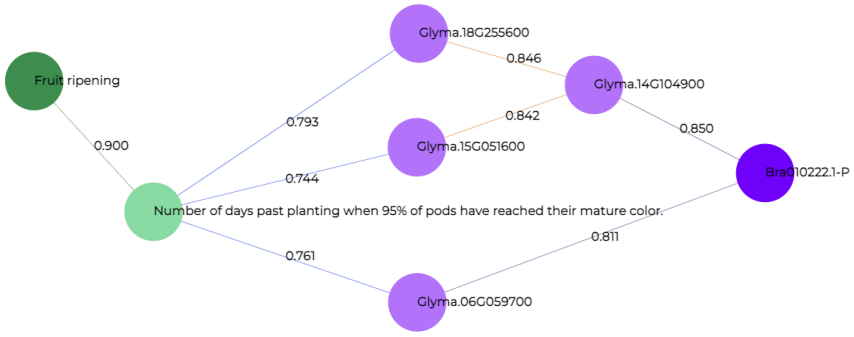


Figure 3.3: Example TGAG

From this "results_table", the TGAG of individual candidate genes can be extracted and displayed on the website.

GeneSieve assigns a score to each candidate gene's TGAG using a specialized type of Graph Neural Network (see Gene ranking models) — higher scores indicating a higher likelihood that the candidate gene affects the input trait.

3.3 Gene ranking models

The rank of a candidate gene within the provided candidate gene set is defined as the score of its TGAG, obtained by passing the TGAG through one of five models that we explore in this thesis.

The five models explored in GeneSieve consist of two Relational-Graph Convolutional Network (R-GCN) models, along with three baseline models, which have all been compared against each other. The inner workings of these models are detailed in the following sections.

3.3.1 Relational-graph convolutional network

The R-GCN models are trained and evaluated using 267 benchmark folders (containing over 3,000 TGAGs in total) that are also viewable on the GeneSieve website. Each folder contains an input trait description and a list of candidate genes. For each candidate gene, GeneSieve constructs a Trait–Gene Association Graph (TGAG) connecting the candidate gene to the input trait through multiple types of biological evidence, as described in Section System Backend and TGAG Construction.

Each TGAG contains four relation types: trait similarity, QTL association, gene co-expression, and gene homology. These relations represent distinct types of biological evidence and may contribute differently to determining whether a candidate gene is associated with the input trait. Consequently, the model must distinguish between relation types rather than treating all edges equivalently. A Relational Graph Convolutional Network (R-GCN) addresses this by learning a separate weight matrix for each relation type, allowing information to be propagated through the TGAG according to the type of biological relationship. This produces node representations that incorporate both graph structure and relation-specific information.

Trait nodes were initialized using DistilBERT embeddings, while gene nodes were initialized using ESM-2 protein embeddings. Four ESM-2 models were evaluated: `esm2_t6_8M_UR50D`, `esm2_t12_35M_UR50D`, `esm2_t30_150M_UR50D`, and `esm2_t33_650M_UR50D`. The `esm2_t30_150M_UR50D` and `esm2_t33_650M_UR50D` models achieved the highest predictive performance and were therefore selected for subsequent evaluation. The resulting trait and gene embeddings were projected to 128-dimensional vectors. The R-GCN hyperparameters were selected through a grid search using an 80–20 train–test split of the 267 benchmark folders. The search evaluated combinations of batch size, number of training epochs, and embedding projection dimension. The opti-

mal configuration was a batch size of 2, 2 training epochs, and a 128-dimensional embedding projection, as shown in Table 3.1.

Table 3.1: Model hyperparameters and evaluated search space.

Hyperparameter	Selected Value	Evaluated Values
Batch size	2	1, 2, 4, 8
Epochs	2	1, 2, 3, 4, 5
Embedding Projection	128	128, 256, 512

GeneSieve explores two R-GCN architectures that use this optimal hyperparameter configuration but differ in how the learned node representations are aggregated and transformed for gene ranking. These architectures were developed to evaluate the effect of different architectural choices on the ranking task. They are referred to as R-GCN Pooled (R-GCN V1) and R-GCN Deep (R-GCN V2). The hyperparameters were initially tuned using the R-GCN Pooled architecture, after which the resulting configuration was also used to train and evaluate the R-GCN Deep architecture. The two R-GCN architectures are detailed in Figures 3.4 and 3.5 below.

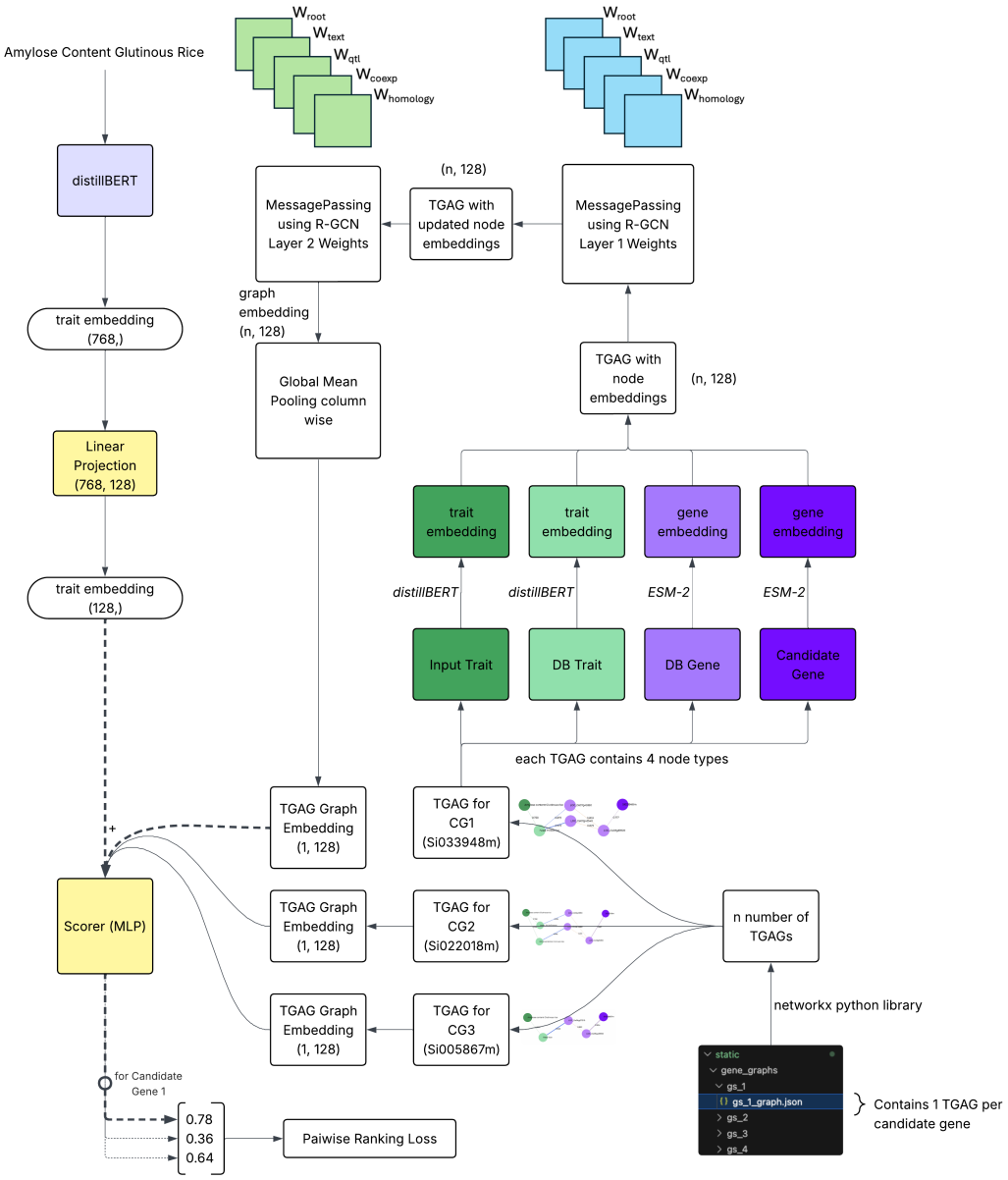


Figure 3.4: R-GCN V1 Architecture

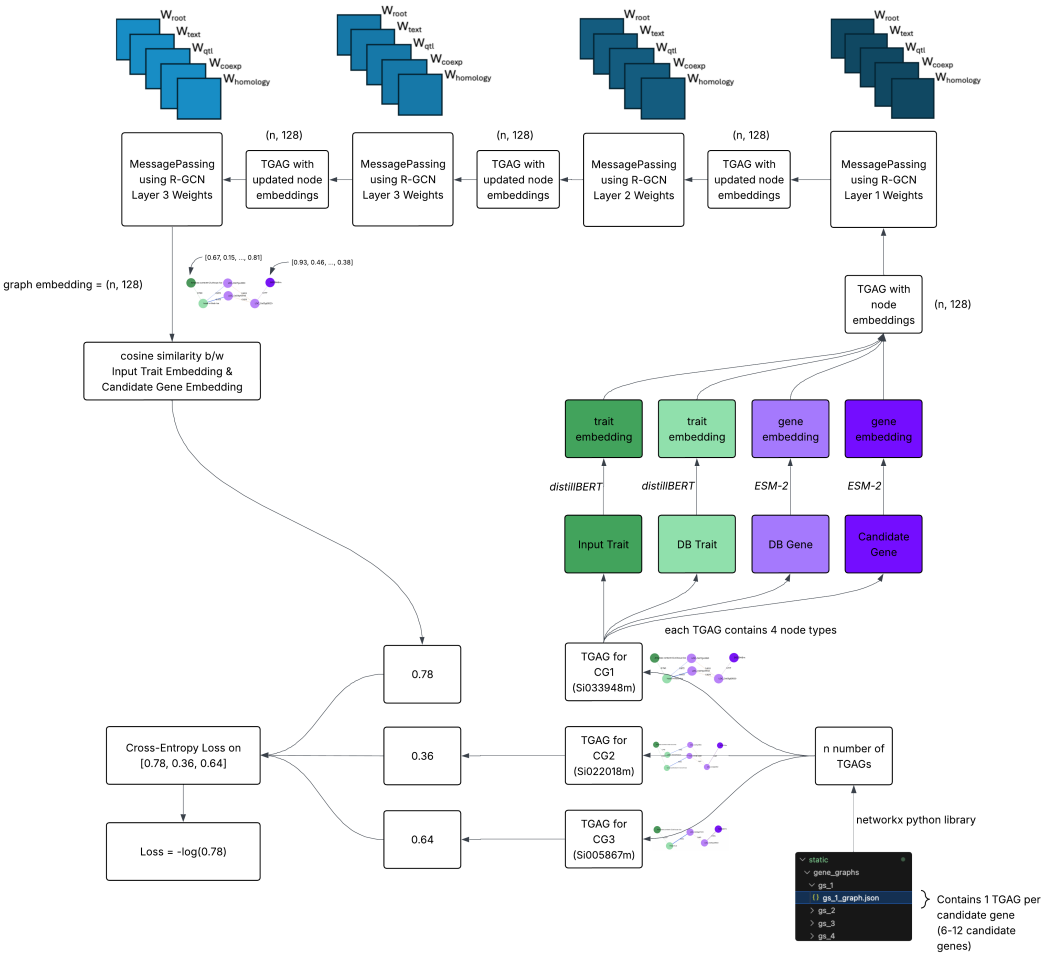


Figure 3.5: R-GCN V2 Architecture

The architecture differences between R-GCN V1 and R-GCN V2 are detailed in Table 3.2. These differences test out the impact of different number of message passing rounds, gene encoders, loss functions, among other model aspects on the model's performance in the gene ranking task.

Table 3.2: Comparison of the two R-GCN architectures.

Component	**R-GCN V1**	**R-GCN V2**
Gene Encoder	ESM-2 t33 650M UR50D	ESM-2 t30 150M UR50D
Gene Embedding Dim	1280	640
Projections	Frozen (to 128-dim)	Learnable (to 128-dim)
Message Passing Rounds	2	4
Edge Weights	Included	Excluded
TGAG Processing	Iterative	Batched
Scoring	Mean pooling + Scorer MLP	Cosine similarity
Loss Function	Pairwise Ranking loss	Cross-Entropy loss
Self Transformation	Double root transform	Single root transform

R-GCN Pooled (V1): This architecture derives a graph embedding of the Trait-Gene Association Graph (TGAG) via global mean pooling after two rounds of message passing. The graph and trait embeddings are concatenated and processed by a Multilayer Perceptron (MLP) to compute a causal link score for gene ranking. The MLP's architecture is detailed as follows: Linear($256 \to 256$) $\to$ LayerNorm $\to$ ReLU $\to$ Dropout(0.2) $\to$ Linear($256 \to 128$) $\to$ ReLU $\to$ Linear($128 \to 1$).

To optimize the model, we employ a pairwise ranking loss designed to maximize the relative score of the true causal gene against all other candidates. Mathematically, this is defined as $L = \frac{1}{|\mathcal{N}|} \sum_{j \in \mathcal{N}} \max(0, 0.3 - (s_{\text{pos}} - s_j))$, where s_{pos} is the true positive gene score, s_j is the score of a non-causal gene, and $\mathcal{N}$ represents the candidate set. Penalizing instances where s_{pos} does not exceed s_j by the margin of 0.3 ensures a uniquely high score for the target gene and a robust decision boundary. This margin hyperparameter was empirically determined to yield the highest inference accuracy among values ranging from 0.1 to 0.8.

R-GCN Deep (V2): In contrast, this deeper architecture employs four rounds of message passing and is optimized using a standard binary cross-entropy loss func-

tion. It streamlines the scoring process by directly computing the cosine similarity between the input trait embedding and the candidate gene embedding to establish the ranking score.

3.3.2 SVD

In GeneSieve, Singular Value Decomposition (SVD) is employed as a latent factor model to derive low-dimensional representations of genes and phenotypes from a heterogeneous biological graph. The graph integrates multiple biological relationship types, including gene homology, gene–trait (QTL) associations, phenotype semantic similarity, and gene co-expression, with weighted edges representing the strength of each relationship. These connections are encoded into a weighted adjacency matrix (A), which is transformed into a symmetrically normalized adjacency matrix

$$\hat{A} = D^{-1/2} A D^{-1/2},$$

where D denotes the diagonal degree matrix. Truncated SVD is then applied to obtain the top ($k = 50$) latent factors, such that

$$\hat{A} \approx U_k \Sigma_k V_k^T,$$

where $U_k \in \mathbb{R}^{n \times k}$ contains the dominant singular vectors and Σ_k contains the corresponding singular values. Because the normalized adjacency matrix is symmetric, the singular vectors capture the dominant spectral structure of the graph. Each node i is represented by the embedding vector

$$\mathbf{u}_i = (U_k)_{i,\cdot} \in \mathbb{R}^{50},$$

which captures its participation across the dominant latent biological patterns encoded within the graph. By embedding both genes and phenotypes into the same

Euclidean latent space, SVD enables the discovery of indirect and multi-hop biological relationships that may not be evident through direct graph traversal alone, while simultaneously reducing the effects of graph sparsity and noise. Consequently, genes and phenotypes that share similar underlying biological connectivity patterns are positioned closer together in the learned latent space, facilitating similarity-based candidate gene ranking.

Gene–trait ranking is performed by comparing the embedding vector of a candidate gene with that of the query phenotype. During inference, the embeddings corresponding to the target phenotype and each candidate gene are retrieved directly from the learned U matrix, and cosine similarity is used to quantify their similarity. Cosine similarity measures the angular alignment between two embedding vectors, providing an indication of whether a gene and phenotype occupy similar regions of the latent space regardless of vector magnitude. Candidate genes are then ranked according to their cosine similarity with the query phenotype, with higher similarity indicating greater proximity in the learned latent space.

3.4 Trait Similarity Models

Attempts to compare phenotypes across studies have largely relied on controlled vocabularies, e.g., the Trait Ontology employed by knowledge bases such as Gramene [5]. Although these approaches have achieved varying degrees of success, their controlled terms cannot encompass the full spectrum of traits and require extensive manual curation and strict adherence to specific terminology. Consequently, a controlled vocabulary or ontology-based approach is impractical for the integrative, cross-species platform that GeneSieve is designed to evolve into. Furthermore, these ontologies substantially constrain the potential to crowdsource results from breeders. GeneSieve leverages advances in large language models (LLMs) to address these chal-

lenges by enabling the system to infer context and meaning from diverse and highly technical vocabulary in the TGAG creation process, reducing the need for rigid term definitions and manual curation.

To compute the similarity between an input trait and a DB trait, we experimented with both a baseline Doc2Vec model and a GPT-4o model (FT-GPT) fine-tuned on a human-annotated dataset. Both the human expert and the models score trait pairs on a scale from 0 (no relation) to 1 (conceptually identical). The fine-tuning of FT-GPT used a grid search over three hyperparameters: number of epochs (1–3), learning rate multiplier (1.2–1.8), and batch size (3–15). Each configuration was evaluated on a 80-20 training-validation split of the human-annotated dataset, with training and validation losses monitored via the OpenAI dashboard. The optimal configuration was 3 epochs, a batch size of 7, and a learning rate multiplier of 1.4.

For evaluating the models, we performed stratified 5-fold cross-validation on the human-annotated dataset, while ensuring proportional representation of input traits across folds despite their uneven distribution. One particular challenge encountered was that the Doc2Vec scores were skewed toward higher values (ranging from 0.67 to 0.96), whereas FT-GPT and human-annotated scores were well spread out in the range from 0 to 1. Therefore it is not meaningful to directly compare the original scores generated by Doc2Vec with that of FT-GPT or the human expert. Furthermore, the human-annotated scores were bimodal (either fairly high or fairly low, with scores in the middle much rarer). This data skewness also makes the comparison of FT-GPT scores and human scores subject to the potential polarized bias of the human expert.

We used two approaches to tackle the score skewness, in order to meaningfully assess the deviation of each model's score from the human-provided score:

1. Using normalized *(norm)* scores: Doc2Vec scores were adjusted to fit a [0, 1] scale, while human-annotated and FT-GPT scores were kept raw. 2. Using percentile *(pct)* scores: The human-annotated, Doc2Vec, and FT-GPT scores were converted into percentiles, with small random perturbations (+/- 0.005) added to scores for resolving ties and ensuring unique percentiles. Given a particular candidate gene, the error of a model is the difference between the model-generated score and human-annotated score, after converting original scores to *pct* or *norm*. Particularly, a 0.1 score difference using the *pct* measure means the candidate gene is ranked 10% higher or lower compared to the human expert's rank.

Table 3.3: Cumulative error across 5 folds using pct and norm scores.

Cumulative Error	F1	F2	F3	F4	F5	Avg
Percentile Scores						
Doc2Vec	0.11	0.11	0.11	0.11	0.11	0.11
Fine-Tuned GPT	0.07	0.07	0.08	0.08	0.07	0.07
Normalized Scores						
Doc2Vec	0.10	0.10	0.10	0.10	0.10	0.10
Fine-Tuned GPT	0.04	0.04	0.05	0.04	0.05	0.05

Table 3.3 reports the measured errors, averaged across all candidate genes within each fold and overall. The results demonstrate that the Doc2Vec model struggles to capture conceptual similarity between plant traits, as evidenced by errors exceeding 0.1, while the FT-GPT model consistently outperforms Doc2Vec across all folds and evaluation measures, achieving up to 54% lower error.

CHAPTER 4

Results and Discussion

Two types of R-GCNs were trained and evaluated on the GeneSieve dataset. There was a process of iterative improvement to the R-GCN custom architecture from an initial prototype (R-GCN V1) to a more refined version (R-GCN V2). Both of their performances have been compared against each other as well as against the three baseline models.

We use two types of comparisons to evaluate the performance of each Gene Ranking Model. The first compares cumulative and non-cumulative performance, while the second compares results obtained by assuming every GeneSieve folder has a length of 20 against results using the actual folder length. These complementary evaluation methods provide different perspectives on model performance. The cumulative comparison captures how well a model ranks the true gene within increasingly larger portions of the candidate list, whereas the non-cumulative comparison highlights performance within specific rank ranges. Similarly, fixing every folder length to 20 enables consistent comparisons across datasets, while using the actual folder lengths reflects the models' performance under real-world conditions where candidate list sizes vary.

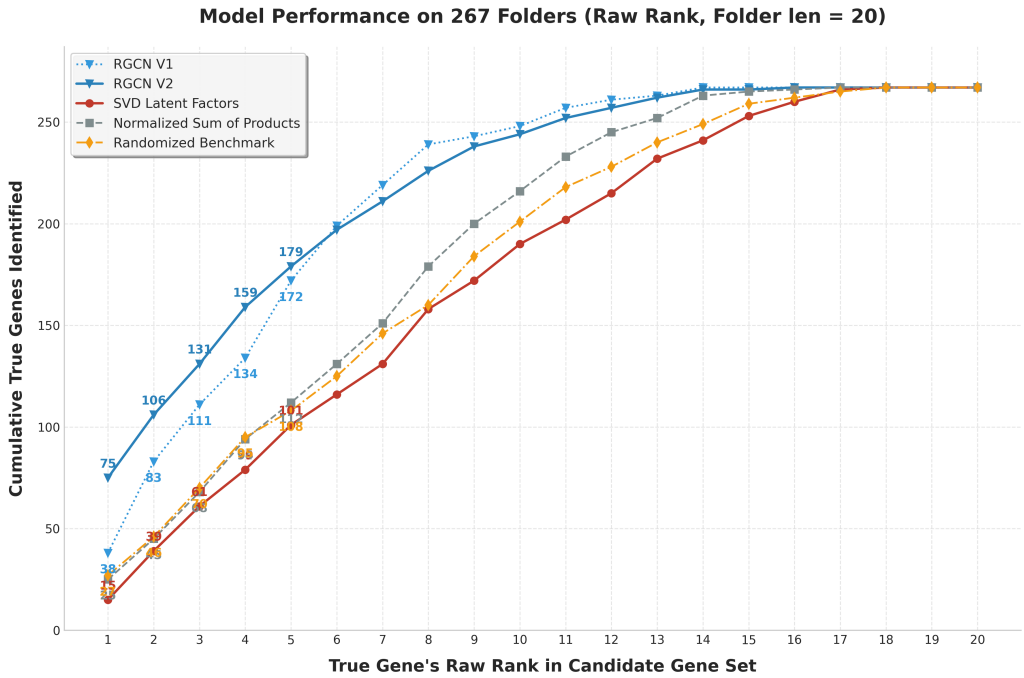


Figure 4.1: **Cumulative Comparison Plot:** Rank is calculated considering folder length = 20

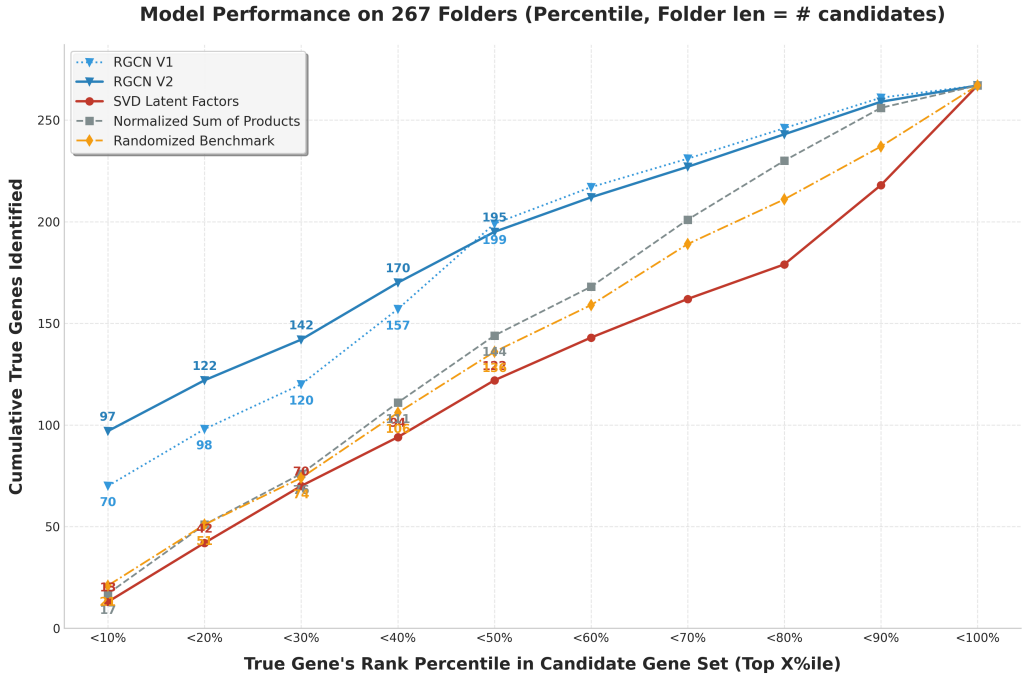


Figure 4.2: **Cumulative Comparison Plot:** Rank is calculated considering folder length = # of Candidate Genes

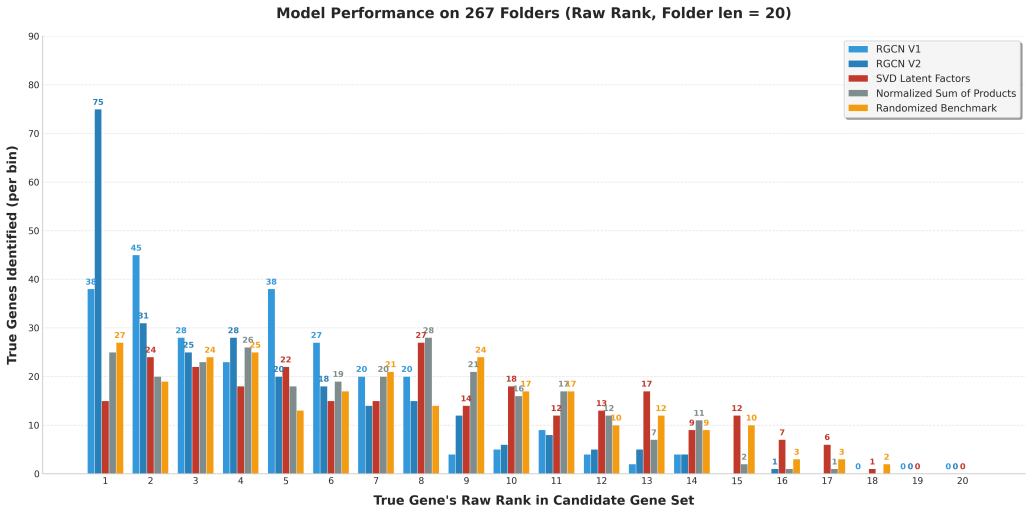


Figure 4.3: **Non Cumulative Comparison Plot:** Rank is calculated considering folder length = 20

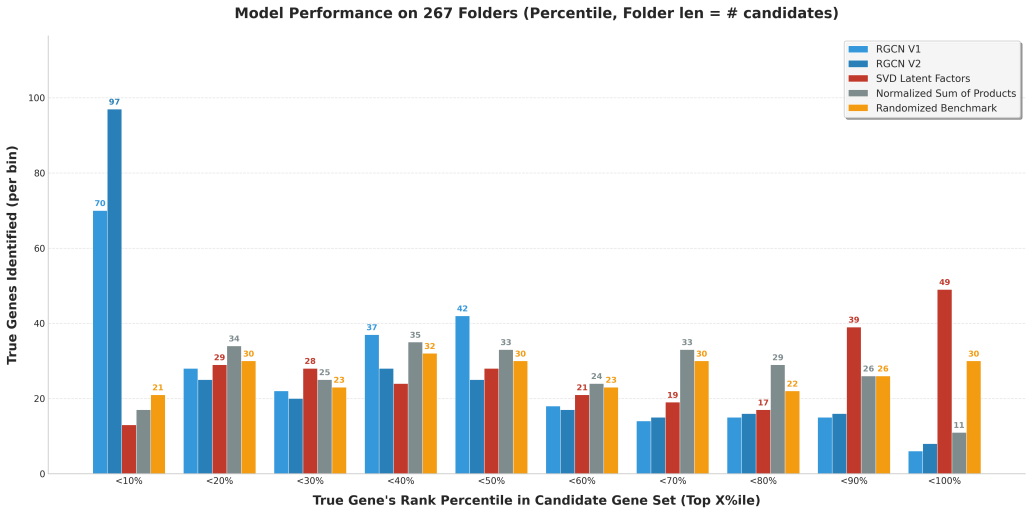


Figure 4.4: **Non Cumulative Comparison Plot:** Rank is calculated considering folder length = # of Candidate Genes

4.1 Relational-Graph Convolutional Network (R-GCN)

As shown in Figures 4.1 and 4.2, both R-GCN architectures substantially outperform the SVD and baseline models. The true gene is ranked within the top 5

candidates approximately 70% of the time, as further demonstrated by the bin-wise results in Figures 4.3 and 4.4.

This improved performance can be attributed to the R-GCN's ability to model the heterogeneous TGAG structure. By learning relation-specific transformations for trait similarity, QTL association, gene co-expression, and homology, the model can integrate multiple types of biological evidence and capture multi-hop relationships between the input trait and candidate genes.

Although both R-GCN architectures substantially outperform the baseline models, their performance differs. One important architectural difference is the number of message-passing rounds. In the TGAG construction used by GeneSieve, the shortest path between the input trait and a candidate gene consists of four hops. Therefore, four rounds of message passing are required for information from the input trait to propagate across this complete path and reach the candidate gene representation. R-GCN Deep uses four rounds of message passing, whereas R-GCN Pooled uses fewer rounds. The stronger performance of R-GCN V2 therefore suggests that allowing information to propagate across the full four-hop path may be beneficial for candidate gene prioritization.

Overall, the results demonstrate that relation-aware graph learning provides an effective approach for integrating heterogeneous biological evidence for candidate gene prioritization.

4.2 SVD

The SVD model demonstrates the lowest performance among the baseline models, suggesting that its simplicity limits its ability to capture meaningful relationships between traits and genes within the TGAGs. This can be attributed to several limitations inherent to the approach.

First, SVD is relation-agnostic. TGAGs are heterogeneous graphs in which four distinct edge types—homology, QTL association, trait text similarity, and co-expression—represent different forms of biological evidence. By representing the graph as a single adjacency matrix, SVD collapses these relation types into a common numerical representation. For example, a homology edge with weight 0.85 is treated identically to a co-expression edge with the same weight, despite their different biological meanings. In contrast, the R-GCN models maintain separate learnable parameters for each relation type, allowing them to learn how different forms of evidence should contribute to gene prioritization.

Second, SVD is unsupervised and its objective is not directly aligned with the ranking task. It does not learn from labeled gene–trait relationships or optimize a ranking objective. Its fixed truncation to the top 50 singular components may also discard sparse, low-variance patterns that are important for identifying causal relationships. Furthermore, SVD is fitted independently to each TGAG, preventing knowledge learned from one graph from transferring to others. Finally, gene–trait scores are computed using fixed cosine similarity rather than a learned scoring function, limiting its ability to capture complex relationships within the heterogeneous graphs.

4.3 Baseline Models

There are two simple models created to serve as a baseline for the SVD and the R-GCN models. The first is what we call the *Sum of Products* model.

In this model, a TGAG's score is taken as the sum of scores of all the individual pathways in it. An individual pathway's score is taken as the product of the weights of the edges on that pathway, where the weights are computed as mentioned in section System Backend and TGAG Construction for the four types of edges.

The scoring formula is below, as diagrammatically depicted in Fig 4.5.

$$\text{TGAG score} = \sum_{i=1}^{n} pathway\ score_i;\ \text{pathway score} = \prod_{j=1}^{m} edge\ weight_j$$

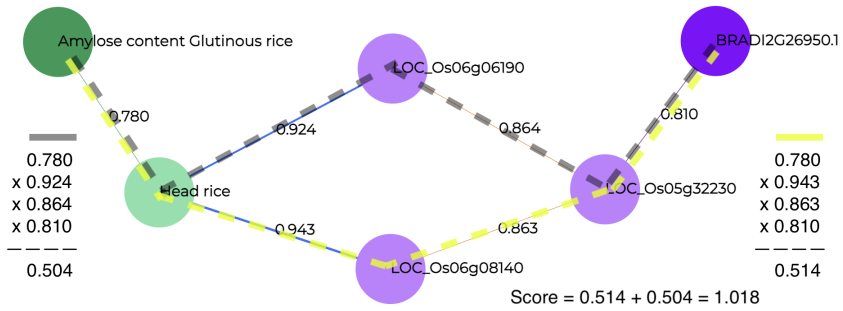


Figure 4.5: **Calculation of the score for a TGAG.** Using summation of products along unique paths connecting the Input Trait and Candidate Gene

This primitive model over-prioritizes large TGAGs containing many nodes and connections. Due to the nature of the algorithm, the numerous pathways between the input trait and a candidate gene within large TGAGs quickly accumulate to produce very large scores. However, a TGAG with more nodes and connections does not necessarily indicate that the candidate gene is more closely associated with the input trait. Consequently, the Sum of Products model serves only as a simple baseline for comparison, as the scoring principle it employs does not accurately reflect biological relevance in plant genomics.

Lastly, we use a purely random baseline model, which we call *"randomize"*. This model randomly selects genes from the candidate gene list without replacement and assigns rankings based on the order in which they are selected. In other words, the first randomly selected gene is assigned rank 1, the second is assigned rank 2, and

so on. When each individual folder is assumed to have a length of 20, this model is expected to produce a flat distribution.

4.4 Trait Similarity Models

Doc2Vec scores are very skewed, and human expert, doc2vec and FT-GPT scores are not distributed in a similar manner. In order to make their results comparable, they must first be converted into the same distribution.

The 3 scores are plotted against their respective normalized score values below.

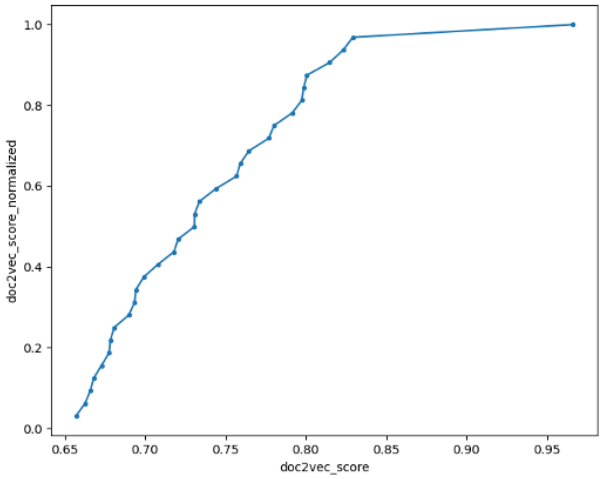


Figure 4.6: **Doc2Vec score normalization.** Original Doc2Vec scores plotted against their normalized scores.

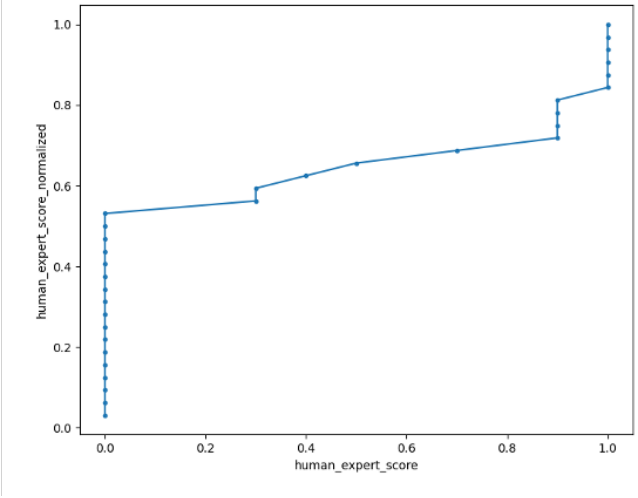


Figure 4.7: **Human expert score normalization.** Original human expert annotation scores plotted against their normalized scores.

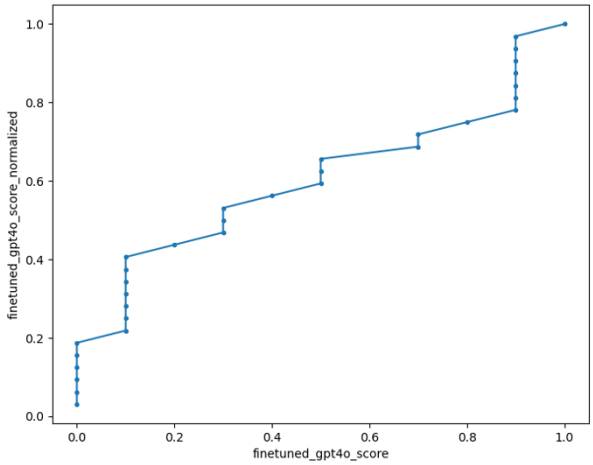


Figure 4.8: **Fine-tuned GPT-4o score normalization.** Original fine-tuned GPT-4o scores plotted against their normalized scores.

Their correlations are also plotted below. From these plots, it is evident that the fine-tuned GPT-4o model performs very well at understanding the human expert's scoring policy as well as picks up on semantic similarity very well, since FT-GPT and human expert scores are highly correlated.



Figure 4.9: **Score Comparison** Doc2Vec (stretched) vs Human Expert Score

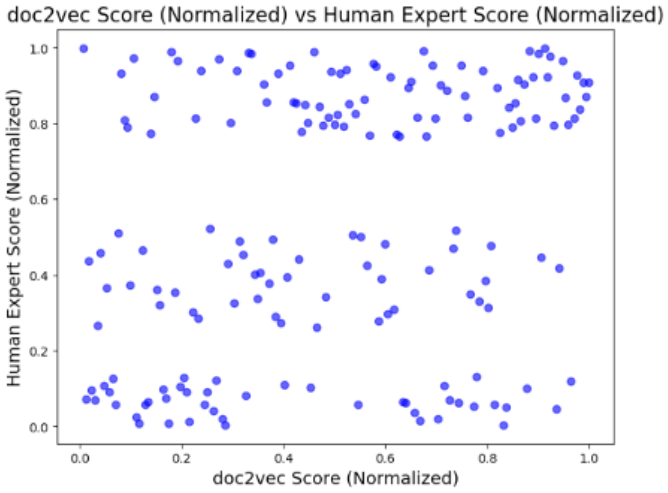


Figure 4.10: **Score Comparison** Doc2Vec (Normalized) vs Human Expert Score (Normalized)

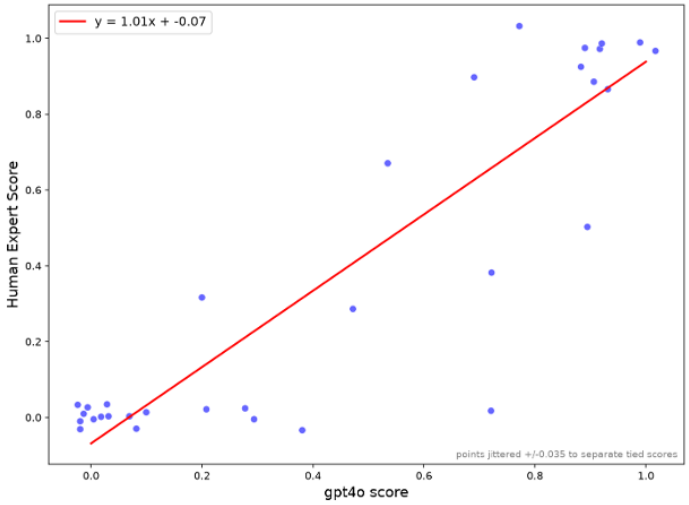


Figure 4.11: **Score Comparison** FT-GPT Score vs Human Expert Score

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

This thesis presents GeneSieve, a framework for candidate gene prioritization that integrates heterogeneous biological evidence through graph learning and large language models. The primary contributions are:

- **Multi-species biological knowledge base and benchmark.** Developed a unified biological knowledge base spanning rice, maize, soybean, and Arabidopsis, linking 1,055 traits to 127,611 genes through 952,630 QTL associations and over 2.4 billion RNA-seq-derived gene co-expression relationships. Additionally, constructed a benchmark dataset of experimentally validated candidate gene prioritization tasks for model training and evaluation.
- **Query-specific Trait-Gene Association Graphs (TGAGs).** Introduced a novel heterogeneous graph formulation that dynamically constructs a graph for each query by integrating LLM-derived trait similarity, curated trait–gene associations, RNA-seq-derived gene co-expression, and sequence homology. Unlike static knowledge graphs or methods relying on isolated evidence sources, TGAGs provide an interpretable, evidence-driven representation tailored to each candidate gene prioritization task.
- **LLM-assisted semantic trait matching.** Fine-tuned a large language model to map natural-language trait descriptions to curated biological traits, overcoming the limitations of exact terminology matching and improving retrieval of relevant biological evidence across species.

- **R-GCN-based candidate gene prioritization.** Designed and evaluated a relational graph convolutional network (R-GCN) that leverages heterogeneous TGAGs to prioritize candidate genes across monocot and dicot species. Experiments on the benchmark dataset demonstrate that the proposed approach consistently outperforms baseline ranking methods, and the complete framework has been deployed as an interactive web platform for the plant science community.

5.2 Future Work

Several directions can further improve the GeneSieve framework. First, additional biological evidence such as protein–protein interaction networks, metabolic pathways, and epigenetic data can be incorporated to enrich TGAGs and provide greater functional context for gene prioritization. Expanding the knowledge base to additional crop species, including wheat, sorghum, and tomato, would also strengthen cross-species inference and improve performance for less-studied plants.

Future work may also explore more advanced multi-modal learning approaches that jointly optimize text, protein sequence, and graph representations. Integrating active learning into the GeneSieve web platform would enable researchers to validate predicted evidence paths and continually improve the model using expert feedback.

Overall, this thesis demonstrates that constructing query-specific heterogeneous graphs and combining them with LLM-driven semantic trait matching and relational graph neural networks provides an accurate and interpretable solution for candidate gene prioritization. By unifying multiple complementary biological evidence sources into a single evidence-driven framework, GeneSieve advances automated gene discovery across diverse plant species and provides a practical tool for accelerating functional genomics and crop improvement research.

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BIOGRAPHICAL STATEMENT

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